

FLORALA, AL, USA

WMO: 720361

Lat: 31.043N

Lon: 86.312W

Elev: 96

StdP: 100.18

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.7	(c) -0.8	(d) -11.4	(e) 1.4	(f) 2.1	(g) -8.5	(h) 1.8	(i) 3.1	(j) 8.9	(k) 21.8	(l) 7.7	(m) 18.8	(n) 1.4	(o) 350	(p) 0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 35.8	(d) 24.6	(e) 34.7	(f) 24.5	(g) 33.6	(h) 24.3	(i) 26.9	(j) 31.6	(k) 26.3	(l) 31.4	(m) 25.7	(n) 30.7	(o) 2.3	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.8	(b) 21.3	(c) 29.0	(d) 25.0	(e) 20.3	(f) 28.4	(g) 24.4	(h) 19.5	(i) 27.9	(j) 85.3	(k) 31.5	(l) 81.9	(m) 31.3	(n) 79.5	(o) 30.8	(p) 31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
7.3	5.8	5.1	DB	-6.0	37.0	2.1	1.6	-7.6	38.2	-8.8	39.1	-10.0	40.0	-11.5	41.2
			WB	-7.2	27.5	2.3	1.1	-8.8	28.4	-10.1	29.0	-11.4	29.7	-13.0	30.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	10.3	12.7	16.2	19.5	23.7	27.1	27.8	27.8	26.0	20.4	14.6	12.6
	DBStd	7.34	5.79	5.13	4.64	3.44	2.88	1.92	1.53	1.78	2.33	4.17	4.47	5.24
	HDD10.0	149	69	29	7	1	0	0	0	0	0	1	10	33
	HDD18.3	889	256	164	97	27	3	0	0	0	0	28	127	187
	CDD10.0	3773	77	107	199	287	424	514	551	551	479	322	149	114
	CDD18.3	1472	6	8	29	63	169	264	293	292	229	91	16	11
	CDH23.3	13626	13	34	216	562	1536	2637	2887	2849	1985	760	117	32
	CDH26.7	5828	0	1	39	128	623	1278	1332	1306	871	232	18	1
Wind	WSAvg	1.9	2.1	2.2	2.3	2.3	1.9	1.6	1.4	1.4	1.6	1.7	1.8	2.0
Precipitation	PrecAvg	1468	134	125	151	119	94	127	154	129	114	81	110	129
	PrecMax	1925	294	298	360	267	233	262	417	257	523	326	303	344
	PrecMin	1024	41	42	16	45	9	46	51	38	25	6	31	46
	PrecStd	214	53	55	83	56	54	60	76	52	93	69	64	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	26.6	29.7	31.2	35.0	37.6	36.4	37.2	35.9	33.1	28.8	25.9
		MCWB	19.2	19.2	19.3	20.5	22.3	23.7	25.0	24.9	23.2	22.8	20.8	20.8
	2%	DB	23.2	24.5	27.7	29.5	33.0	35.8	34.9	35.4	34.1	30.9	26.3	24.2
		MCWB	19.0	18.6	20.1	20.2	22.3	24.5	24.9	25.4	23.5	22.1	19.8	19.9
	5%	DB	21.6	22.6	25.8	28.1	31.5	34.1	33.8	34.1	32.8	29.4	24.4	22.8
		MCWB	18.3	17.7	18.8	19.4	22.0	24.3	24.9	25.2	23.6	21.2	18.9	19.6
	10%	DB	19.7	21.1	23.9	26.8	30.2	32.9	32.7	32.8	31.5	27.6	22.7	21.0
		MCWB	16.9	17.2	17.5	18.7	21.6	24.1	24.7	24.9	23.5	20.3	17.8	18.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	20.9	25.5	23.6	26.4	28.7	29.3	27.9	26.1	25.4	23.0	22.6
		MCDB	22.7	23.6	27.1	27.0	28.8	32.4	31.2	33.4	29.9	28.6	25.8	24.0
	2%	WB	20.2	20.2	21.1	22.3	24.0	26.1	26.6	26.9	25.5	24.3	21.9	21.2
		MCDB	22.2	22.8	24.2	26.1	29.0	31.9	31.7	32.1	30.1	27.5	24.3	23.1
	5%	WB	19.0	19.2	20.1	21.6	23.3	25.4	25.9	26.4	25.0	23.3	20.3	20.0
		MCDB	21.0	21.8	23.4	25.3	28.4	31.2	31.3	31.3	29.7	26.6	23.2	21.9
	10%	WB	17.3	17.9	19.0	20.6	22.8	24.9	25.5	25.8	24.5	22.2	18.9	18.9
		MCDB	19.6	20.4	22.6	24.3	27.8	30.4	30.5	30.5	29.0	25.8	22.0	20.8
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.2	12.7	12.8	12.3	11.1	10.3	10.0	10.7	12.4	13.4	11.0
		MCDBR	11.3	12.3	13.3	13.9	13.8	13.0	11.6	11.7	12.1	12.8	13.2	10.9
		MCWBR	7.5	7.7	6.8	6.0	5.0	4.2	3.8	4.0	4.1	5.4	7.7	7.4
	5% WB	MCDBR	9.6	10.4	11.3	10.7	11.2	11.3	10.2	10.2	9.8	10.0	10.9	9.8
		MCWBR	7.5	7.8	7.9	6.1	5.1	4.5	4.2	4.0	3.9	5.3	7.8	7.9
Clear-Sky Solar Irradiance	taub	0.325	0.336	0.359	0.384	0.431	0.465	0.515	0.499	0.432	0.373	0.346	0.331	
	taud	2.532	2.505	2.435	2.371	2.271	2.235	2.107	2.151	2.313	2.454	2.473	2.526	
	Ebn at Noon	906	927	924	908	862	829	786	795	839	873	871	879	
	Edh at Noon	86	96	111	123	136	141	160	151	124	100	90	83	
All-Sky Solar Radiation	RadAvg	3.01	3.60	4.86	5.91	6.49	6.18	5.92	5.51	5.05	4.42	3.41	2.63	
	RadStd	0.21	0.46	0.40	0.48	0.41	0.50	0.36	0.38	0.35	0.47	0.30	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (45 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+100

Nomenclature: See separate page